

5. (2 points)

A Cape Cod loss reserving calculation has the following inputs and estimates:

- Total premium is \$10,000,000.
- Estimated ELR is 65%.
- Process variance/mean ratio is 50,000.
- The parameter covariance matrix is:

	ELR	ω	θ
ELR	0.0029	-0.0042	0.19
ω	-0.0042	0.0055	-0.41
θ	0.19	-0.41	25.52

a. (1 point)

Calculate the coefficient of variation of prospective losses.

b. (0.5 point)

Briefly describe what process variance and parameter variance of the prospective losses measure.

c. (0.5 point)

Briefly describe whether the Cape Cod method typically has a higher or lower parameter variance than the chain ladder method.

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SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

QUESTION 5	
TOTAL POINT VALUE: 2.0	LEARNING OBJECTIVE(S): A2, A3
SAMPLE ANSWERS	
Part a: 1.0 point	
<u>Sample 1</u> $E[Loss] = ELR \times Premium = 10,000,000 \times 0.65 = 6,500,000$ $Process\ Variance = \sigma^2 \times E[Loss] = 50,000 \times 6,500,000 = 3.25 \times 10^{11}$ $Parameter\ Variance = Var[ELR] \times Premium^2 = 0.0029 \times 10,000,000^2 = 2.9 \times 10^{11}$ $Std\ Dev = \sqrt{Process\ Var. + Parameter\ Var.} = \sqrt{3.25 \times 10^{11} + 2.9 \times 10^{11}} = 784,219$ $CV = \frac{Std\ Dev}{E[Loss]} = \frac{784,219}{6,500,000} = 12.06\%$ <u>Sample 2</u> $Mean = 10M \times 0.65 = 6.5M$ $\frac{Process\ Var}{Mean} = 50,000; Process\ Var = 3.25 \times 10^{11} = Process\ Std\ Dev = 570,088$ $CoV_{process} = \frac{Process\ Std\ Dev}{Mean} = \frac{570,088}{6,500,000} = 0.0877$ $CoV_{parameter} = \frac{\sqrt{0.0029}}{0.65} = 0.0828$ $CoV = \sqrt{CoV_{process}^2 + CoV_{parameter}^2} = \sqrt{0.0877^2 + 0.0828^2} = 0.121$	
Part b: 0.5 point	
<u>Sample Responses</u> • Process variance measures the uncertainty that arises due to random fluctuations in the data, while parameter variance measures uncertainty due to parameter estimates differing from the true parameters of the distribution. • Process variance: risk from inherent randomness of the insurance process. Parameter variance: risk from the incorrect parameterization of all predictors. • Process variance: randomness inherent in the insurance modeling process. Parameter variance: uncertainty arising from parameter selection/estimation. • Process variance is natural uncertainty due to inability to predict the insurance process. Parameter variance is uncertainty in the parameters used to estimate prospective losses.	
Part c: 0.5 point	
<u>Sample Responses</u> • Cape Cod has a lower parameter variance since it is using more information (premium exposure). • Cape Cod method has lower parameter variance as less parameters are being estimated.	

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

- Cape Cod usually has a lower parameter variance because LDF method overparameterizes the data. Cape Cod uses less parameters. Cape Cod also incorporates more information (premium/exposure) which also leads to a lower parameter variance.

EXAMINER'S REPORT

Part a

Candidates were expected to understand the relationship that the process and parameter variances had to the coefficient of variation of prospective losses. Additionally, they were expected to be able to calculate both variances and the expected value of prospective losses based on the information provided, and then correctly combine them for the total CV.

Common mistakes included:

- Using an incorrect equation for parameter variance (e.g. trying to calculate the parameter variance of reserves instead of prospective losses).
- Using the correct parameter variance equation but replacing the premium with expected losses.

Part b

Candidates were expected to be able to properly define both process and parameter variance in sufficient detail.

A common mistake included defining parameter variance as “the variance of the parameters”, instead of mentioning that it is a result of the estimated/selected parameters.

Part c

Candidates were expected to understand the differences in parameters between the Cape Cod and the chain ladder methods, and how the differing number of parameters influences the size of the parameter variance.

A common mistake included correctly identifying that the Cape Cod method has fewer parameters than the chain ladder method, but then stating that this equated to a higher parameter variance.